

Hybrid Deep Reinforcement Learning for UAV Inspection in Large-Scale Wind Farms: Deployment and Routing Optimization

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Abstract—The growth of global wind capacity necessitates efficient maintenance and inspection, particularly in offshore wind farms. However, existing manual inspection method is time consuming and lacks scalability. This article introduces an autonomous unmanned aerial vehicle (UAV) inspection method for offshore wind farms by combing heuristic algorithm with deep reinforcement learning (DRL). Our approach integrates UAVs deployment and routing optimization, factoring in environmental conditions, such as wind and weather. The inspection task is divided into UAVs deployment and routing optimization, with models constructed for UAVs, offshore wind farms, and wind conditions. A heuristic algorithm is used to optimize UAVs deployment, ensuring efficient coverage despite the UAVs' limited range. For routing optimization, we develop a DRL model for effective path planning of UAVs simultaneously. Experimental results demonstrate the superior efficiency of our approach in terms of coverage, energy consumption, and adaptability to environmental changes. By introducing the hybrid DRL algorithm, a scalable and robust solution for offshore wind farm inspections is developed, enhancing both operational efficiency and reliability.

Index Terms—Deep reinforcement learning (DRL), multi-object optimization, routing optimization, unmanned aerial vehicle (UAV), wind farm inspection.

I. INTRODUCTION

AS GLOBAL energy demand continues to grow and environmental concerns become more prominent, wind energy

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has become an important renewable energy resource to achieve the carbon neutrality target. In Q1 2023, the global installed wind capacity reached 23.5 GW, marking a significant increase year-over-year of 27%, and it is expected that the cumulative global installed wind power capacity will surpass the remarkable milestone of 1 TW by the end of 2023 [1]. Considering that wind turbines (WT) are exposed to outdoor elements, they are susceptible to various operational threats. Critical components of a WT, such as the gearbox, generator, and bearings, have proven to be particularly vulnerable, experiencing frequent failures [2]. The inspection and maintenance of turbines pose significant challenges for large-scale wind farms, particularly for the offshore wind farm scenarios. In the process of turbine inspection and maintenance, the importance of data collection and fault detection methods cannot be overstated [3]. For example, in [4], a model for estimating blade icing conditions is proposed by collecting data on turbine blade icing conditions. However, the challenge of collecting pertinent data, implementing appropriate fault detection algorithms in offshore wind farms, and positioning sensors should not be underestimated. In such scenarios, the use of unmanned aerial vehicles (UAVs) for data acquisition and automated inspections emerges as a viable solution [5].

Given the limitations of UAVs' battery capacity, effective path optimization becomes a critical concern [6], [7]. Liu et al. [8] created intelligent UAV trajectory optimization based on sampled points. Van Huynh et al. [9] proposed low-complexity convex optimization algorithms for planning UAV flight paths to minimize flight time. Tran et al. [10] proposed the dynamic programming algorithm, which achieves a balance between the speed and quality of the solution in order to minimize the total energy consumption while performing the task. Sun et al. [11] obtained optimal 3-D trajectories through monotonic optimization while reducing energy consumption. Al-Hilo et al. [12] used proximal policy optimization to solve the UAV trajectory optimization and resource allocation problem. Nguyen et al. [13] proposed a deep reinforcement learning (DRL) algorithm to plan the shortest flight path for UAVs, while maximizing data collection from IoT devices. Fu et al. [14] propose a two-tier decision-making architecture to determine optimal routes for UAVs assigned to inspect grid damage. However, the above literature only utilized a single UAV to complete the task, also these methods are limited for large scale area.

TABLE I
SUMMARY OF RELATED WORKS

Paper	Deployment optimization	Trajectory planning	Environmental impact	Scalability
[8], [9], [10], [11], [12], [13], [14]	×	✓	×	×
[15]	×	✓	✓	×
[16], [17], [18]	✓	×	×	×
[19], [20], [21], [22], [23], [24], [25]	✓	✓	×	×
[26], [27]	✓	✓	✓	×
Ours	✓	✓	✓	✓

In the aspect of UAV deployment, Huang et al. [15] proposed a differential evolutionary algorithm based on the problem of optimizing the deployment of UAVs, where the size of the population can be dynamically varied. Chen et al. [16] proposed a comprehensive scheme that can minimize the number of UAV deployments while maximizing the data transfer rate. Ngoenriang et al. [17] presented a heuristic algorithm to estimate the cost of UAV flights and solve the problem of UAV deployment. Zhang et al. [18] proposed a population-based three-layer structural optimization algorithm to solve the UAV deployment and flight planning problem. Han et al. [19] proposed an improved dandelion algorithm to simultaneously optimize the number and location of UAV deployments and UAV flight trajectories. Ji et al. [20] adopted the block alternating descent and successive convex approximation methods to solve the deployment and trajectory optimization problem. Ji et al. [21] considered both static and dynamic UAV deployments, then proposed a spiral algorithm to optimize the UAV trajectory. Al-Habob et al. [22] proposed three improved genetic algorithms (GA) to solve the combinatorial optimization problem for UAV deployment as well as trajectory planning. Sobouti et al. [23] used the finite state machine (FSM) algorithm to solve the UAV placement problem; then they proposed a binary linear problem model to find the optimal trajectory. Chai et al. [24], [25] enhanced the feasibility and efficiency of constrained trajectory optimization by incorporating noise terms into the actor network's sampling weights within the DDPG framework while improving multi-objective particle swarm optimization (MOPSO's) bias selection and local search mechanisms. Chung et al. [26] proposed a wind model that takes into account the importance of wind in the UAV offshore wind farm inspection problem. Then, they have transformed the problem of UAV inspection in wind farms into two distinct problems: the UAV placement problem and the routing problem. Finally, they designed heuristic algorithms to solve the problem. However, this work sets the maximum number of UAV inspections as a constant value. This setting may not be optimal for different wind farm configurations, as it does not consider the dynamic nature of UAV battery capacity and turbine distribution.

A summary of the related work which investigated the UAV deployment and routing optimization problem is presented in Table I. However, most research addresses land-based situations, with the deployment and routing of UAVs in offshore wind farms being less studied. At the same time, there is less related work that simultaneously investigates both UAV deployment and routing optimization methods while considering environmental

impacts. To better solve the problems mentioned above and inspired by distributed computing approaches which considering both energy consumption, battery capacity, and communication range [27], [28], [29], we propose a DRL-based autonomous UAV inspection algorithm for large-scale offshore wind farms that takes into account both UAV deployment and routing optimization. Detailed contributions are introduced as follows.

- 1) A heuristic algorithm-based UAV deployment method that incorporates environmental wind conditions and UAV operational constraints to optimize inspection sub-areas, improving coverage and deployment efficiency compared to traditional geometric partitioning.
- 2) We formulate UAV routing optimization as an asymmetric traveling salesman problem (ATSP) under wind disturbances and develop a DRL algorithm trained on subdivision data to generate energy-efficient inspection paths. The DRL solver demonstrates superior computational efficiency and adaptability over conventional methods, reducing energy consumption.
- 3) A combination of the heuristic deployment algorithm and the DRL-based routing optimization model, offering a scalable solution suitable for different offshore wind farm configurations. The proposed hybrid method can adapt to varying numbers of turbines and wind farm sizes, enhancing the operational efficiency and reliability of wind farm inspections.

The rest of this article is organized as follows. Section II formulates the research problem, including region delineation and path planning. Section III introduces the proposed heuristic algorithm for region delineation and the DRL model for path planning. Section IV presents the experimental design, relevant parameters, and comparisons with related methods. Finally, Section V concludes this article and suggests possible directions for future research.

II. SYSTEM MODEL AND PROBLEM FORMULATION

In this section, we comprehensively consider the requirements of the UAV inspection task and construct the wind farm model, the wind model and the UAV model, respectively. The wind farm inspection problem can be divided into two subproblems: 1) UAVs deployment problem; and 2) UAV routing optimization for each subarea.

A. System Model

The offshore wind farm model comprises the coordinates of the Q turbines and the wind farm's topology. The distance between turbine m and turbine n is $d_{m,n}$. In order to construct a detailed and practical model of the inspection UAV, we consider four important factors that have a large impact on offshore wind farms, including the flight, load, wind, and altitude.

1) *Flight*: This study considers both the UAV's airspeed V_a and groundspeed V_g . V_a is vital for lift and aerodynamics, while V_g impacts mission time and coverage. In windy conditions, a headwind requires a higher V_a to maintain V_g , increasing energy use; tailwinds have the reverse effect. Different airspeed profiles also affect the UAV's maneuverability and stability during gusts

or turbulence. The UAV's flight time is T_f and its flight power is P_f . The flight power consumption of the UAV inspection is calculated

$$E_f = P_f T_f. \quad (1)$$

UAVs need to hover to take pictures of turbine blades, gearboxes, generator sets and other equipment to gather information. If the UAV hovers for time T_h and the hovering power is P_h , the energy consumption for hovering is calculated as

$$E_h = P_h T_h. \quad (2)$$

2) Load: The total weight of the UAV consists of its airframe mass and the payload, which usually includes sensors like high-resolution visible light cameras, infrared thermal imagers, and light detection and ranging (LiDAR) systems. These onboard sensors enable capturing details such as blade surface cracks and corrosion, and detecting abnormal thermal signatures or hot spots. An increase in payload linearly raises the flight power requirement and simultaneously reduces the UAV's maneuverability, maximum climb rate, and wind resistance. In addition, payload distribution is crucial for flight stability, as incorrect mass distribution can cause control issues or loss of control.

3) Wind: In tailwind conditions, a UAV can achieve the same V_g with a lower V_a , conserving energy, extending endurance and operational range. Conversely, headwinds require a higher V_a to maintain V_g , increasing motor power and energy consumption, reducing the operational radius and endurance. Wind gusts, which are short-intense wind speed pulsations, can disrupt a UAV's attitude stability despite its ability to withstand high average wind speeds, risking loss of control. Thus, defining safety thresholds based on wind speed, gust intensity, and turbulence is essential. $0, \frac{\pi}{2}, \pi$ and $\frac{3\pi}{2}$ represent the wind speed V_w in the north, east, south, and west directions, respectively. The corresponding wind data will also change depending on the altitude at which the UAV is flying. The UAV ground speed is given

$$V_a + V_w = V_g. \quad (3)$$

4) Altitude: Determining UAV inspection flight altitude requires considering multiple factors. Different sensors have optimal ranges: high-resolution cameras need close proximity for detail, while wide-angle thermal imagers can scan from afar. So, altitude must meet the needs of all onboard sensors. For safety, the UAV must stay clear of turbine blade rotation areas and follow local flight altitude regulations. In addition, as altitude rises, lower air density affects flight endurance.

B. UAV Deployment Problem

One of the key objectives of this study is to minimize the costs associated with UAV inspections while ensuring the successful completion of inspection tasks. In particular, optimizing the quantity of inspection UAVs is essential, as it directly corresponds to the number of UAV nests deployed. As described in [30], UAV nests serve as convenient hubs for UAVs to recharge or swap batteries. In addition, these nests are capable of overseeing UAV inspection operations and processing the data gathered by the UAVs. Consequently, the operational range

of UAVs during inspection missions is influenced by both the UAVs' inherent flight range and their communication distance capabilities R_c . To ensure that the UAV can safely return to the UAV nest after the inspection mission, the furthest distance R_f^w between the UAV and the UAV nest should be set to half the maximum flight distance of the UAV. The maximum flight distance is determined by the combination of the maximum flying speed $V_{\max}^w$ and the flying time $T_{\max}$ of the UAV i in the wind W . The maximum inspection range R_i^w of the UAV i at altitude h and wind W can be expressed as

$$R_i^w = \min(R_c, R_f^w) \quad (4)$$

$$\mathbf{R} = [x - x_r, y - y_r] \quad (5)$$

$$R_f^w = \frac{V_{\max}^w T_{\max}}{2}. \quad (6)$$

After the definition of the scope of the UAV inspection work, the UAV deployment problem can be formulated as follows:

$$\min_{B, Z, x_i, y_i} \sum_{i=1}^N b_i \quad (7a)$$

$$\text{s.t } b_i, z_{i,m} \in \{0, 1\} \quad \forall i \quad (7b)$$

$$\sum_{i=1}^B z_{i,m} = 1 \quad \forall i, m \quad (7c)$$

$$\sum_{i=1}^B z_{i,m} \leq M \quad \forall i, m \quad (7d)$$

$$\sum_{m=1}^M d_{m,m+1} + d_{M,1} \leq 2R_f^w \quad (7e)$$

$$E_f + M E_h \leq E_{\max} \quad (7f)$$

$$\sum_{m=1}^Q z_{i,m} = 1 \quad \forall m \quad (7g)$$

$$[x_m, y_m] \in R_i^w \quad \forall z_{i,m} = 1 \quad (7h)$$

$$[x_i, y_i] \in \{[x_m, y_m]\} \quad \forall z_{i,m} = 1 \quad (7i)$$

where $\mathbf{B} = [b_i]_{1 \times U}$ and $\mathbf{Z} = [z_{i,m}]_{U \times Q}$ representing the set of UAV nests and the set of inspection areas respectively. B represents whether the UAV nests are deployed or not, if $b_i = 1$ it represents the deployment of the UAV nests, otherwise it represents the removal of the UAV nests. Matrix $\mathbf{Z}$ represents the inspection area collection, which can also be considered as the UAV set. It stores the numbers and coordinates of all the turbines in the divided inspection area. If turbine m is inspected by UAV i , then $z_{i,m} = 1$, otherwise $z_{i,m} = 0$. Constraint (7a) defines the optimization objective as the deployment of the minimum number of UAV nests to complete the offshore wind farm inspection task. The optimal number of UAV nests to deploy leads to cost optimization. In constraint (7b), b_i and $z_{i,m}$ are defined. Constraint (7c) states that all UAVs cannot belong to more than one UAV nest. Constraint (7d) states that a UAV inspects at most M turbines, where M is determined by the battery capacity of

the UAV, which ensures that the UAV completes the inspection task. Constraint (7e) gives a solution for M . Constraint (7f) states that the sum of the UAV's flight energy consumption and hover energy consumption cannot exceed the UAV's maximum battery capacity $E_{\max}$. Constraint (7g) states that each turbine is inspected by one and only one UAV. Constraint (7h) states that all turbines in the delineated UAV inspection subarea must be within the UAV flight range. Constraint (7i) states that the deployment location of the UAV nest must be the location of the turbines within the UAV inspection area.

C. UAV Inspection Routing Optimization Problem

After dividing the UAV inspection area, we need to study the UAV inspection path planning problem in the area. Z_i denotes the area inspected by UAV i , containing the number and coordinates of the turbine to be inspected. The topology of the inspected area can be obtained from the acquired coordinates. The UAV path planning problem can be considered as a UAV i that starts from the UAV nest, inspects all the turbines and finally returns to the UAV nest. Thus, this problem can be formulated as follows:

$$\min \sum_{m=1}^M \sum_{n \in Z} t_{i,m,n} \quad (8a)$$

$$t_{i,m,n} = \frac{d_{mn}}{V_{g,m,n}^w} \quad (8b)$$

$$t_{i,n,m} = \frac{d_{mn}-0}{V_{g,n,m}^w} \quad (8c)$$

$$\sum_{i \in U} \sum_{m \in Z} x_{i,m,n} \geq 1 \quad \forall m, n \in Z \quad (8d)$$

$$\sum_{i \in U} x_{i,m,n} = \sum_{i \in U} x_{i,n,m} \quad \forall m, n \in Z \quad (8e)$$

$$\sum_{m \in Z} (x_{i,m,n} + x_{i,n,m}) \geq 1 \quad \forall m, n \in Z \quad (8f)$$

$$\sum_{m=1}^M \sum_{n \in B} t_{i,m,n} \leq T_{\max} \quad \forall n \in Z. \quad (8g)$$

The optimization objective of path planning is to minimize inspection time. Due to the presence of wind, the flight time of the UAV i from turbine m to n and from turbine n to m must be calculated as in (8b) and (8c), respectively. Constraint (8d) ensures that each turbine is inspected by the UAV. Constraint (8e) ensures that the UAV completes its inspection mission from the UAV nest and finally returns to the UAV nest. Constraint (8f) indicates that any turbine must be patrolled at least once. Constraint (8g) ensures that the planned inspection path of the UAV has to be limited by the maximum endurance for the time of the inspection.

III. METHODOLOGY

As shown in Fig. 1, we investigate an offshore UAV inspection system where a UAV flies from a deployed nest to each WT

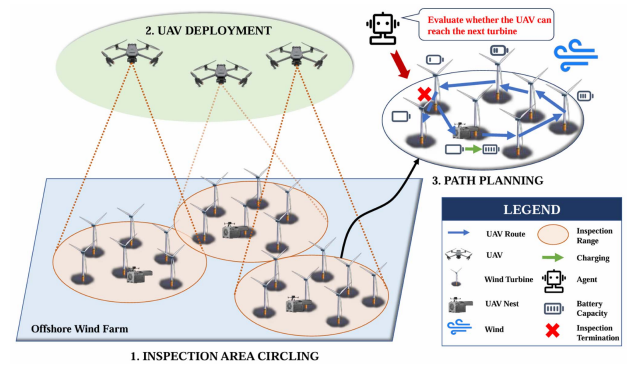


Fig. 1. Overview of UAV inspections in offshore wind farms.

to collect relevant data. The proposed hybrid DRL approach combines a heuristic algorithm for UAV deployment with a DRL-based path planning algorithm. The heuristic algorithm provides a practical and quick solution for deployment, while the DRL algorithm enables real-time adaptation and optimization of the inspection paths, resulting in a more efficient and robust UAV inspection system.

1) *Stage 1—Data Preprocessing*: In this stage, the primary focus is on preprocessing the data related to the wind farm. The layout information of the wind farm is first obtained, and the Mercator projection method is employed to convert latitude and longitude coordinates into Cartesian coordinates, resulting in the wind farm topology. Using the UAV's maximum wind resistance specifications, the allowable wind speed range for inspection is analyzed.

2) *Stage 2—Inspection UAVs Deployment*: In this stage, the inspection areas are divided based on the UAV's battery capacity and the inspection requirements of the wind farm. Each inspection area is assigned a UAV nest to allow the UAV to recharge. The heuristic algorithm is designed to solve the large-scale problem of UAV deployment in offshore wind farms. It takes into account factors, such as UAV range, battery capacity, and turbine distribution. By using a heuristic approach, we can quickly find a near-optimal deployment solution that divides the wind farm into inspection areas.

3) *Stage 3—Inspection Routing Optimization*: In this stage, the inspection path is planned for the divided inspection areas. The DRL-based path planning algorithm then focuses on optimizing the path within each of these inspection areas. It learns from the environment, including wind conditions and turbine locations, to find the most efficient path for the UAV to visit all the turbines in the area and return to the nest. This combination allows us to leverage the strengths of both methods. Considering the influence of wind, the path planning problem is transformed into an ATSP. A graph pointer network is trained as a local model, which takes wind information and inspection area division results as input to quickly generate the inspection path.

A. Heuristic Algorithm-Based UAVs Deployment Method

The UAVs deployment method proposed in this article is divided into three steps (shown in Fig. 2). The first step is to obtain

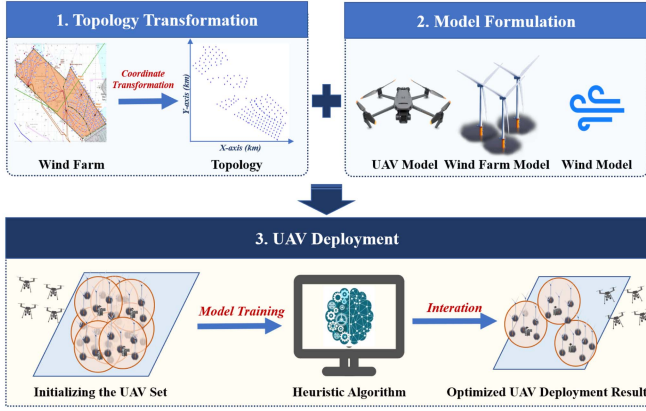


Fig. 2. Block diagram of inspection UAV deployment.

the topology of the wind farm and initialize the deployment of one UAV at each turbine location, i.e., initialize the deployment of Q UAVs. The second step is to obtain the UAV flight range and UAV battery capacity based on the constructed UAV model, wind farm model and wind model. In the third step, the maximum number of UAV inspections is determined using a heuristic algorithm. The number of UAV deployment is then optimized. In order to carry out inspection zoning of offshore wind farms for UAVs, it is necessary to clarify the size of the inspection area for the UAVs. According to (6), the radius of the UAV inspection area is taken as the minimum of the communication distance and the maximum inspection distance. As the communication distance is a constant, we only need to analyze the maximum flight distance of the UAV i under the effect of wind w .

With known flight range, all turbine numbers within the flight range of the UAV can be placed in A by obtaining the turbine and UAV coordinates. Then we need to build the feasible group by removing some redundant connections between the UAV and the turbine. Specifically, we need to ensure that the number of turbines in all UAV sets does not exceed M . M is a variable determined by constraint (7e), which ensures that all UAV sets are legal. The detailed process is provided in Algorithm 1.

Construction of viable groups for UAV inspections by 1) initialization of multiple groups, 2) calculating flying and hovering energy consumption, and 3) checking that the total energy consumption of the UAV does not exceed the maximum battery capacity. After constructing the viable group, we obtained the feasible group with the same number of turbines. It is clear that although all feasible groups are legitimate, the number of UAV deployments is not optimal. To optimize, if all turbines in a feasible group are covered by other groups, that group is removed. For the remaining repeatedly inspected turbines, the connection to the nearest UAV is retained, and redundant connections are eliminated. This ensures each turbine is inspected only once, thereby optimizing UAV deployment and reducing costs. The proposed UAVs deployment method differs from the approach in [26] in two main aspects. First, we determine the maximum number of UAV inspections based on UAV battery capacity and turbine distribution in a more flexible way, while [26] sets it as a constant. In a wind farm with turbines spread

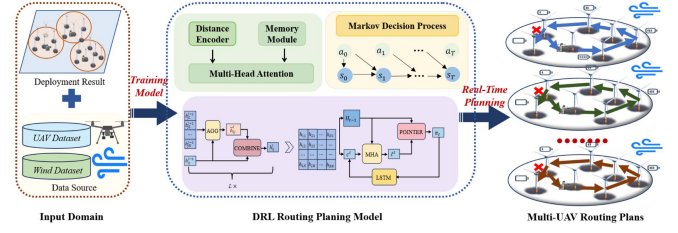


Fig. 3. Block diagram of inspection path planning.

Algorithm 1: Viable Group Construct and UAV Nest Number Minimization.

Input: The coordinates of Q turbines, $R_i^w, E_{max}, P_f, T_f, E_h, B, Z$
Output: $Z_{new}, x_i, y_i, B_{new}$

```

1 Set  $U = Q$ , set every  $[x_i, y_i]$  to  $[x_m, y_m]$ 
2 for  $i = 1$  to  $U$  and  $m = 1$  to  $Q$  do
3   if  $[x_i, y_i] \in R_i^w$  then
4      $z_{i,k} = 1$ 
5   else
6      $z_{i,k} = 0$ 
7   for  $i = 1$  to  $U$  do
8     Sort turbines in a decreasing order based on distance
      between their location and UAV nest as  $g_1, g_2, \dots, g_{|U_i|}$ 
9      $m = g_1$ 
10    if  $P_f T_{g_1, g_2} + E_h + P_f T_{g_2, g_1} \leq E_{max}$  then
11       $path = \{g_1, g_2\}$ 
12    else
13      remove  $g_2, g_3, \dots$  from  $U_i$ 
14  Repeat 8
15 for  $i = 1$  to  $Z$  do
16    $(Z_i \cap Z_1) \cup (Z_i \cap Z_2) \cup \dots (Z_i \cap Z_m) = Union\{i\}$ 
17   while  $Z_i \subseteq Union\{i\}$  do
18      $s_i = 0, delete = True;$ 
19     for  $m = 1$  to  $Q$  do
20       if  $m \in Z_m, m \in Z_{m+1}, \dots, m \in Z_n$  then
21          $Inter(m) = \{Z_m, Z_{m+1}, \dots, Z_n\}$ 
22         Calculate  $Z_m$  in  $Inter(m)$  based on distance between
          turbine  $m$  and UAVs as  $d_{m,1}, d_{m,2}, \dots, d_{m,n}$ 
23     if delete then
24       for  $i = 1$  to  $Z$   $m = 1$  to  $Q$  do
25         if  $d_{m,i} < d_{m,j}$  then
26            $z_{i,m} = 0$ 
27         else
28            $z_{j,m} = 0$ 
29        $step = step + 1$ 
30   Repeat

```

over a large area and UAVs with limited battery capacity, our method can adjust the number of UAVs and their inspection areas to ensure complete coverage. Second, our DRL-based routing optimization algorithm (introduced in Section III-B) takes into account more environmental factors, such as wind direction and speed variations at different altitudes, resulting in more efficient paths.

B. DRL-Based UAV Routing Optimization Algorithm

As shown in Fig. 3, after dividing the wind farm into smaller inspection areas, the DRL model then focus on optimizing paths within these local regions, making the optimization process more tractable. In each subarea, the nest is not only the departure

point of UAV but also providing the charging facilities, and a UAV depart from the UAV nest and inspect each turbine, final return to the UAV nest to recharge. The object of UAV is to find the optimal routing within the subarea by considering environment changes. Given the topology of all subareas, UAV sets and weather conditions, DRL-based routing planning model is trained to find the optimal routes for all UAVs. The Markov decision process of DRL-based routing optimization algorithm can be defined as a tuple (S, A, R, P) , consisting of state S , action A , reward R , and state transition P , which are introduced in detail as follows.

1) **State Space S** : In the context of UAV inspection of WT, the state $s_t \in S$ at a decision-making time step t is defined as a composite vector. This vector is designed to comprehensively encapsulate the UAV's current operational environment and its intrinsic status, facilitating optimal path planning decisions. Specifically, s_t comprises the following key components.

- a) *Set of visited turbines (V_t)*: $V_t = \{v_1, v_2, \dots, v_k\}$, where v_i denotes the identifier of a turbine that has already been inspected. This information is crucial for avoiding redundant visits and ensuring complete coverage of all target turbines.
- b) *Current position and remaining energy (L_t, E_t)*: $L_t = (x_t, y_t)$ represents the UAV's coordinates in a Cartesian space. E_t quantifies the UAV's remaining battery energy or flight endurance. These parameters directly constrain the selection of subsequent reachable turbines and the overall mission sustainability.
- c) *Information on unvisited turbines (U_t)*: $U_t = \{u_1, u_2, \dots, u_m\}$, where each $u_j = (\text{pos}_j, \text{dist}_j)$ encapsulates data for unvisited turbine j . This includes its geographical position pos_j and its Euclidean distance dist_j from the UAV's current location.
- d) *Environmental information (W_t)*: This component includes critical meteorological data, pertinent to the UAV's operation, such as the average wind speed w_{avg} , gust intensity w_{gust} , and wind direction w_{dir} at the current location or target region. Thus, $W_t = (w_{\text{avg}}, w_{\text{gust}}, w_{\text{dir}})$. This information is vital for assessing energy consumption and flight risks associated with navigating to the next turbine.

Consequently, the state s_t can be formulated as $s_t = [V_t, L_t, E_t, U_t, W_t]$.

2) **Action Space A** : At each decision epoch t , given the current state s_t , the agent (UAV) selects an action a_t from the action space A . In this turbine inspection task, the action a_t is defined as the selection of the next turbine to be inspected. The action space is discrete and can be represented as

$$a_t \in U_t(s_t) \cup \{v_{\text{depot}}\} \quad (9)$$

where $U_t(s_t)$ is the set of currently unvisited turbines derived from state s_t , and v_{depot} represents the action of returning to the UAV nest. This latter action is typically chosen when remaining energy is critically low or all designated turbines have been inspected. If a specific turbine $u_j \in U_t(s_t)$ is selected, then $a_t = u_j$. The selection of a_t is governed by a policy $\pi(a_t|s_t)$, which aims to choose actions that maximize the expected long-term cumulative reward (or minimize long-term cumulative cost).

3) **State Transition P** : The state transition dynamics describe how the system evolves from the current state s_t to the next state s_{t+1} after the agent executes action a_t . This transition can be modeled by a function $P(s_{t+1}|s_t, a_t)$. The components of the state are updated as follows.

- a) *Set of visited turbines update*: If action $a_t = u_j$ (i.e., selecting turbine u_j for inspection), the set of visited turbines is updated as $V_{t+1} = V_t \cup \{u_j\}$.
- b) *Current position update*: The UAV's position L_{t+1} becomes the coordinates of the selected turbine pos_j (if $a_t = u_j$) or the coordinates of the depot $\text{pos}_{\text{depot}}$ (if $a_t = v_{\text{depot}}$).
- c) *Remaining energy update*: The remaining energy E_{t+1} is calculated as $E_{t+1} = E_t - E_{\text{consumed}}(L_t, L_{t+1}, W_t)$. Here, E_{consumed} is an energy consumption model that estimates the energy expended based on the flight distance (determined by L_t and L_{t+1}) and the prevailing wind conditions W_t .
- d) *Information on unvisited turbines update*: The set of unvisited turbines is updated to $U_{t+1} = U_t \setminus \{u_j\}$. Furthermore, for all turbines remaining in U_{t+1} , their respective distance attributes (dist) must be recalculated based on the UAV's new position L_{t+1} .

While parts of the state transition are deterministic given the action, the overall transition may be stochastic if environmental factors like wind (W_t) are modeled probabilistically. For planning purposes, expected values or robust estimates (e.g., worst-case scenarios for energy consumption) are often used.

4) **Reward Function R** : The objective is to minimize the time cost, and the penalty function is defined as follows. To guide the agent toward learning a policy that minimizes the total time cost of the inspection tour, the reward function $r(s_t, a_t, s_{t+1})$ provides an immediate scalar feedback signal after executing action a_t in state s_t and transitioning to state s_{t+1} . Our primary objective of minimizing flight distance translates to a principal reward component that is the negative of the distance flown. Additional auxiliary rewards and penalties can be incorporated to shape more desirable behaviors. The immediate reward R_t at time step t is formulated as

$$R_t = w_d \cdot R_{\text{dist}} + w_e \cdot R_{\text{energy}} + w_c \cdot R_{\text{completion}} + w_s \cdot R_{\text{safety}} \quad (10)$$

where the coefficients w_d, w_e, w_c , and w_s are nonnegative weights that balance the contribution of each component to the total reward. The agent's objective is to learn a policy π that maximizes the expected sum of discounted future rewards $G_t = \sum_{k=0}^{T-t-1} \gamma^k R_{t+k+1}$, where $\gamma \in [0, 1]$ is the discount factor. The rewards and penalties are introduced as follows.

- a) *Distance cost reward (R_{dist})*: This component directly penalizes the distance traveled

$$R_{\text{dist}}(s_t, a_t, s_{t+1}) = -d(L_t, L_{t+1}) \quad (11)$$

where $d(L_t, L_{t+1})$ is the Euclidean distance between the previous location L_t and the new location L_{t+1} resulting from action a_t . This negative reward encourages the selection of shorter path segments.

- b) *Energy consumption penalty* (R_{energy}): This penalizes the energy consumed during the transition

$$R_{\text{energy}}(s_t, a_t, s_{t+1}) = -E_{\text{consumed}}(L_t, L_{t+1}, W_t). \quad (12)$$

This term promotes energy-efficient flight, particularly under adverse wind conditions.

- c) *Task completion reward* ($R_{\text{completion}}$): Positive rewards are given for progressing in the mission. A reward C_{visit} is given upon successfully visiting a designated turbine u_j : $R_{\text{completion}} = C_{\text{visit}}$ if $a_t = u_j \in U_t(s_t)$. A larger terminal reward C_{final} is awarded when all specified turbines have been inspected and the UAV safely returns to the depot.
- d) *Safety and constraint violation penalties* (R_{safety}): Penalties are applied for undesirable or unsafe behaviors. This term can be a sum of various penalties P_k : 1) $P_{\text{low_energy}}$: A penalty if E_{t+1} falls below a predefined safety threshold E_{min} before reaching the depot. A substantial penalty is incurred if energy is depleted mid-mission. 2) P_{revisit} : A penalty if action a_t leads to revisiting a turbine already in V_t (if not explicitly allowed by the strategy for other reasons). 3) $P_{\text{unsafe_flight}}$: A penalty if the UAV operates or continues to operate when environmental conditions W_t exceed predefined safety thresholds.

The coordinates of each WT are feature coded using an encoder to provide a global correlation of all the turbines. The input sequence X_i is upgraded to a d_x -dimensional high-dimensional embedding vector $\bar{X}_i$ by the embedding layer. Each input node is characterised by $X_i = [x_{xi}, x_{yi}]$, where $\mathcal{X}_{xi}$ and $\mathcal{X}_{yi}$ are the horizontal and vertical coordinates of the client node. Global node features are obtained using multihead attention (MHA) modules, each of which consists of a MHA layer and a feedforward layer. Each attention layer is processed using skip-connection and batch normalization

$$\tilde{h}_i = \text{Normalization}^l \left(h_i^{(l-1)} + MHA_i^l(h_1^{(l-1)}, \dots, h_n^{(l-1)}) \right). \quad (13)$$

The attention features of the l th layer are obtained from the features of the $l-1$ st layer

$$h_i = \text{Normalization}^l \left(\tilde{h}_i + FF^l(\tilde{h}_i) \right). \quad (14)$$

The MHA mechanism can obtain the entire graph on the information about the node weights of the whole graph. The value of the node information depends on the relevance of the key and the query

$$q_{i,n} = W_n^Q h_i \quad (15)$$

$$k_{i,n} = W_n^K h_i \quad (16)$$

$$v_{i,n} = W_n^V h_i \quad (17)$$

where $W_n^Q \in \mathbb{R}^{d_k \times d_h}$, $W_n^K \in \mathbb{R}^{d_k \times d_h}$, and $W_n^V \in \mathbb{R}^{d_v \times d_h}$. The correlation $\mathcal{U}_{ij}^n$ can be computed based on $q_{i,n}$ and $k_{i,n}$, and the correlation of two nonadjacent nodes i and j is $-\infty$. The softmax activation function is applied to the correlation $\mathcal{U}_{ij}^n$ to obtain the attention weight $-a_{ij}^n$, where $a_{ij} \in [0, 1]$. The hidden vector h'_i of node i can be obtained by performing a scaling dot

product operation on $\mathcal{V}_j^n$ from the computed attention weight of the point. Finally, multihead feature fusion is performed to obtain the final node features, and W_n^0 is the network parameter of the n th head

$$u_{ij}^n = \begin{cases} \frac{q_{i,n}^T k_{j,n}}{\sqrt{d_{k,n}}}, & \text{if } i \text{ adjacent to } j \\ -\infty, & \text{otherwise} \end{cases} \quad (18)$$

$$a_{ij}^n = \text{softmax}(u_{ij}^n) \quad (19)$$

$$h'_{i,n} = \sum_j a_{ij}^n \times v_j^n \quad (20)$$

$$\text{MHA}_i(h_1, \dots, h_n) = \sum_{n=1}^N W_n^O h'_{i,n}. \quad (21)$$

The feedforward network consists of two linear layers, the first layer is connected to the second layer after the neurons have been activated by rectified linear unit (ReLU), where $W^{ff,0}, b^{ff,0}, W^{ff,1}, b^{ff,1}$ is the fully connected network parameter of the first and second layers

$$FF(h'_i) = W^{ff,1} \cdot \text{ReLU}(W^{ff,0} \hat{h}_i + b^{ff,0}) + b^{ff,1}. \quad (22)$$

The decoder uses an attention layer and pointer vectors, and uses the softmax layer to generate the next client node vector. The attention mechanism and the pointer vector are defined as follows, where W_r and W_q are trainable parameters

$$u_i^{(j)} = \begin{cases} v^T \cdot \tanh(W_r r_j + W_q q), & \text{if } j \neq \pi(k), \forall k < j \\ -\infty, & \text{otherwise.} \end{cases} \quad (23)$$

The encoder uses the fan information h_0 at the t th decoding step as the query vector q for the attention mechanism, and the output hidden vector h_i for all client nodes as the context vector r_j . For nodes that have already been accessed, a mask masking operation is used so that the node cannot be accessed subsequently. All alternative client nodes are derived on the basis of the policy network

$$\pi_\theta(a_i | s_i) = p_i = \text{softmax}(u_i). \quad (24)$$

Selection of the next client node $a_i = X_{i+1}$ by sampling or greedy search. The reinforcement algorithm based on the band baseline is used for learning. After feeding the data into the network, the output is generated as a probability distribution $\pi_\theta(a_i | s_i)$ and gradient descent is used to optimize the generated probability distribution. The gradient descent at the k th layer can be written as

$$\begin{aligned} \nabla_{\theta_k} J(\theta_k) = & \frac{1}{B} \sum_{i=1}^B \left[\left(\sum_{t=1}^N r_k \left(s_{i,t}^{(k)}, a_{i,t}^{(k)} - b_{i,k} \right) \right. \right. \\ & \left. \left. \times \sum_{t=1}^N \nabla_{\theta_k} \log \pi_{\theta_k}(a_{i,t}^{(k)} | s_{i,t}^{(k)}, h_{i,t}^{(k)}) \right) \right]. \end{aligned} \quad (25)$$

Updating rules by gradient descent

$$\theta_k \leftarrow \theta_k + a \nabla_{\theta_k} J(\theta_k). \quad (26)$$

TABLE II
DETAILED DATA OF WALNEY OFFSHORE WIND FARM

Offshore wind farm	Coverage area(km ²)	Turbine numbers	Generationl capacity(MW)
Walney 1	28	51	183.6
Walney 2	45	51	183
Walney 3	149	87	659
Walney	222	189	1025.6
Walney500	587	500	2712
Walney1000	1175	1000	5423
Walney2000	2350	2000	10846

TABLE III
DETAILED PARAMETERS OF DJI MATRICE 30

Description	Parameter	Value
Carrying capacity	$L_{\max}$	4069 g
Wind resistance	V_w^R	12 m/s
Maximum flight speed	$V_{\max}$	23 m/s
Maximum flight time	$T_{\max}^f$	41 min
Maximum hovering time	$T_{\max}^h$	36 min

IV. CASE STUDY

In order to validate the practicality of the method proposed in Section III, a large offshore wind farm called Walney offshore wind farm, which consists of three subfarms is selected as the dataset (shown in Table II). The scalability of the proposed algorithm is also examined, using the three Walney wind farm datasets as a baseline, a Gaussian kernel density estimation was used to fit the real turbine distribution to generate a virtual turbine distribution. Through proportional scaling of the wind farm dimensions, we constructed virtual wind farms containing 500, 1000, and 2000 turbines, designated as Walney 500, Walney 1000, and Walney 2000 wind farms. For the UAV, DJI Matrice 30 is choose for simulation testing. The UAV is equipped with binocular vision, near-infrared sensors and laser rangefinders for all-round obstacle avoidance. It also supports mapping. The maximum operating radius of the DJI UAV Nest adapted to this UAV is 7 km. The parameters and values of the UAV are given in Table III. The energy consumption for hovering is approximately 1.14 times the energy consumption for flying. With each turbine hovering for 10 s for testing, the energy expended can be considered to have flown for 11.4 s. In addition, wind data from Walney Island was used in the modeling. The average wind speed on Walney Island is shown in Fig. 4. According to [26], the percentage of values where the quotient of hourly gusts and hourly mean wind speeds is less than 2 for Walney Island is about 93.14%. This means that by selecting $\bar{V}_w = 6$ m/s, the probability of gusts exceeding $V_w^R = 12$ m/s does not exceed 7%. Therefore, choosing $\bar{V}_w = 6$ m/s can ensure that the UAV safely completes inspection tasks.

A. Results of UAVs Deployment in Wind Farm

The proposed heuristic algorithm is tested in Walney wind farms. The specific iteration process is shown in Fig. 5. Initially, 87 UAVs are deployed at the Walney 3 wind farm and a feasible group construction is performed. After 30 rounds of iteration

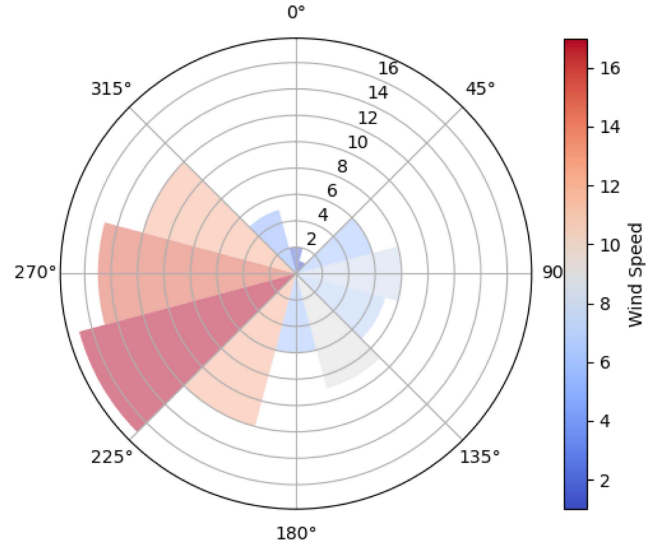


Fig. 4. Walney Island wind rose.

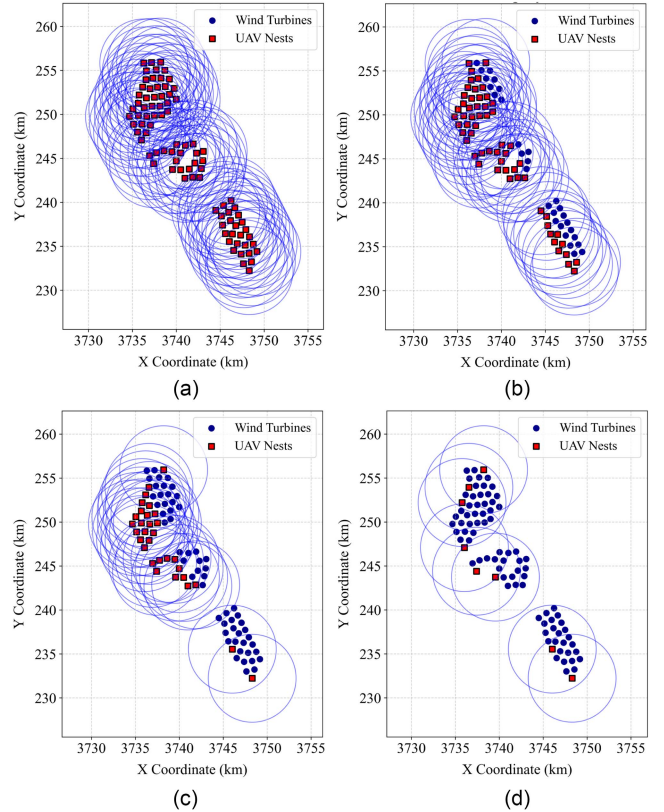


Fig. 5. UAVs deployment process in Walney 3 wind farm. (a) Step = 0, B = 87. (b) Step = 30, B = 58. (c) Step = 60, B = 31. (d) Step = 90, B = 8.

using the heuristic algorithm, 58 UAVs remain on the wind farm. After 90 rounds of iteration, only eight UAVs need to be deployed in the wind farm while the iteration is convergency. Table IV compares UAV deployment numbers across different offshore wind farms with varying turbine counts under different wind speeds for four algorithms: [26], ant colony optimization

TABLE IV
UAV NUMBERS OF UAV DEPLOYMENT IN WIND FARMS

Offshore wind farm			Walney 1	Walney 2	Walney 3	Walney	Walney 500	Walney 1000	Walney 2000
Turbine numbers			51	51	87	189	500	1000	2000
Algorithm	Wind Speed (m/s)	M	Number of UAVs						
[26]	6	5	15	13	24	61	138	273	550
		10	8	9	12	42	77	149	301
	8	5	15	13	24	63	138	273	550
		10	8	9	12	54	77	149	301
ACO	6	5	11	11	21	47	127	262	545
		10	6	6	11	24	66	146	303
	8	5	12	12	22	48	134	264	545
		10	6	7	11	25	70	150	307
GA	6	5	11	12	22	50	137	271	537
		10	7	7	11	25	68	143	295
	8	5	12	12	23	54	143	280	550
		10	7	7	12	27	72	159	308
Ours	6	\	4	4	8	19	56	82	158
	8	\	4	4	9	20	60	85	160

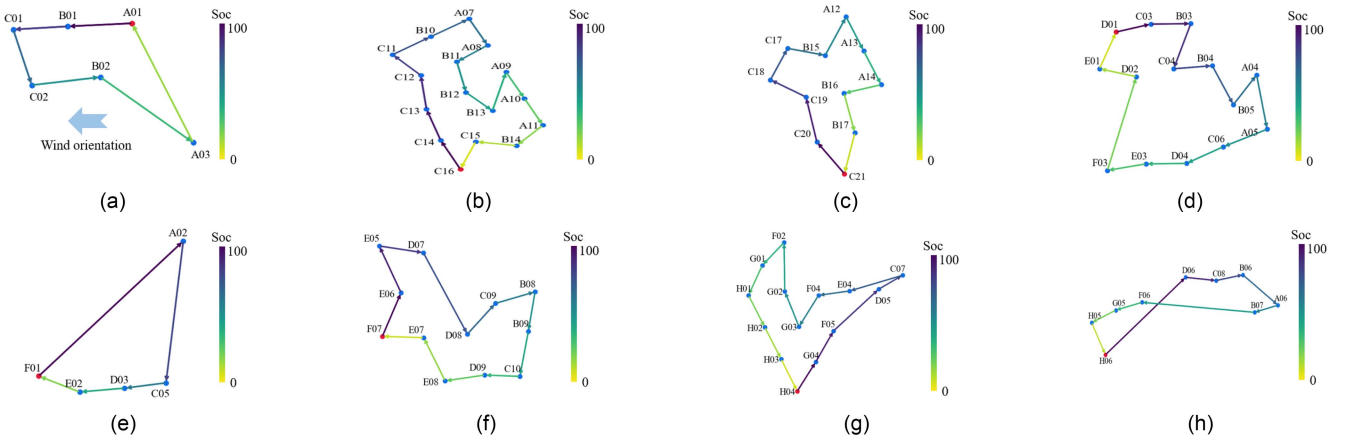


TABLE V
PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS ON WIND FARM DATASETS

Algorithm	Walney 1			Walney 2			Walney 3		
	Flight Time	Run Time	Gap	Flight Time	Run Time	Gap	Flight Time	Run Time	Gap
	(min)	(s)	(%)	(min)	(s)	(%)	(min)	(s)	(%)
Brute-Force	11.8794	351.8842	0	9.6566	5350.6859	0	11.8833	5178.7768	0
Branch-and-Bound	11.8794	8.9706	0	9.6566	12.6161	0	11.8833	55.7988	0
OR-Tools	12.6411	0.1383	6.41	9.9279	0.1445	2.81	12.1392	0.1363	2.15
Greedy	17.0749	0.0010	43.74	11.6760	0.0261	20.91	13.6866	0.0010	15.18
[26]	11.8794	0.8504	0	9.6566	2.9231	0	11.8833	17.0945	0
Ours	11.8794	0.0356	0	9.6566	0.0430	0	11.8833	0.3831	0

that get the optimal results have a running time that exceeds our algorithm compared to other algorithms. The ones with faster running time than ours do not get the optimal results. Therefore, our proposed DRL model gets the optimal results while reducing the running time.

V. CONCLUSION

This article decomposes the inspection problem of large-scale offshore wind farms into two sequential stages: area division and path planning. To address these two problems, we construct a comprehensive model that incorporates UAVs, offshore wind farms, and wind conditions. A heuristic algorithm was introduced to manage UAV deployment in large-scale offshore wind farms, addressing UAVs' limited range. In addition, a DRL model was developed for optimizing path planning, enabling efficient navigation across multiple inspection subareas. Our experimental design and comparative analysis demonstrated the effectiveness of our methods in terms of coverage efficiency, energy consumption, and adaptability to environmental changes. The heuristic algorithm for UAV deployment and the DRL-based path planning model together offer a scalable solution applicable to various offshore wind farm configurations. Future research could explore real-time adaptation strategies, enhance model robustness against extreme conditions, and integrate real-time fault detection algorithms.

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